

☒ EASY

☐ ALGEBRA

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# Algebra

10 questions · ~15 min

# 17

**Q1****ALGEBRA**

If  $4x - 28 = -24$ , what is the value of  $x - 7$ ?

A. -24

B. -22

C. -6

D. -1

A student council group is selling school posters for a fundraiser. They use the function  $px = 5x - 220$  to determine their profit  $px$ , in dollars, for selling  $x$  school posters. In order to earn a profit of \$ 900, how many school posters must they sell?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$x = 10$$

$$y = x + 21$$

The solution to the given system of equations is  $x, y$ . What is the value of  $y$ ?

- A. 2.1
- B. 10
- C. 21
- D. 31

Q4

GRID-IN ALGEBRA

The equation  $46 = 2a + 2b$  gives the relationship between the side lengths  $a$  and  $b$  of a certain parallelogram. If  $a = 9$ , what is the value of  $b$ ?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

**Q5**

ALGEBRA

$$4x + 6 = 18$$

Which equation has the same solution as the given equation?

A.  $4x = 108$

B.  $4x = 24$

C.  $4x = 12$

D.  $4x = 3$

**Q6**

ALGEBRA

For the linear function  $f$ , the graph of  $y = fx$  in the  $xy$ -plane has a slope of  $\frac{1}{4}$  and passes through the point  $(0, 5)$ . Which equation defines  $f$ ?

A.  $fx = \frac{1}{4}x + 5$

B.  $fx = \frac{1}{4}x + \frac{1}{5}$

C.  $f(x) = \frac{1}{4}x - \frac{5}{4}$

D.  $fx = \frac{1}{4}x - 5$

Angela is playing a video game. In this game, players can score points only by collecting coins and stars. Each coin is worth  $c$  points, and each star is worth  $s$  points.

- The first time she played, Angela scored 700 points. She collected 20 coins and 10 stars.
- The second time she played, Angela scored 850 points. She collected 25 coins and 12 stars.

Which system of equations can be used to correctly determine the values of  $c$  and  $s$  ?

- A.  $10c + 20s = 700$   
 $12c + 25s = 850$
- B.  $20c + 10s = 700$   
 $25c + 12s = 850$
- C.  $20c + 700s = 10$   
 $25c + 850s = 12$
- D.  $700c + 20s = 10$   
 $850c + 25s = 12$

$$3a + 4b = 25$$

A shipping company charged a customer \$25 to ship some small boxes and some large boxes. The equation above represents the relationship between  $a$ , the number of small boxes, and  $b$ , the number of large boxes, the customer had shipped. If the customer had 3 small boxes shipped, how many large boxes were shipped?

- A. 3
- B. 4
- C. 5
- D. 6



Q9

GRID-IN

ALGEBRA

$$6x + k = 6x + 5$$

In the given equation,  $k$  is a constant. If the equation has infinitely many solutions, what is the value of  $k$ ?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A bus is traveling at a constant speed along a straight portion of road. The equation  $d = 30t$  gives the distance  $d$ , in feet from a road marker, that the bus will be  $t$  seconds after passing the marker. How many feet from the marker will the bus be 2 seconds after passing the marker?

- A. 30
- B. 32
- C. 60
- D. 90